

OUTPUT:

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● vjti-comp@VJTI-COMPS-iMac-4 FINAL_SE_LAB % g++ book_ride_full.cpp
● vjti-comp@VJTI-COMPS-iMac-4 FINAL_SE_LAB % ./a.out
=====
Book Ride - Ride Sharing Application
Student : Manasa Gopisetty | ID: 241071039 | Batch C
Course : Software Engineering (R5IT2009T)
VJTI Mumbai | End Semester Lab Exam | 30-05-2026
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RIDE SHARING APP - DEMO SIMULATION
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[Booking 1] Login_rider: Manasa Gopisetty | Vehicle=SEDAN | Matunga -> Bandra
Fare Estimate : Rs.115.73
Distance : 5.32 km
[Dashboard] Ride #1001 | Status: PENDING | Fare: Rs.115.73 | Vehicle: Sedan
Driver_decision Assigned : Ramesh Kumar (MH01AB0001)
Status : IN_PROGRESS ... riding ...
=====
RIDE RECEIPT
=====
Ride ID : 1001
Rider : Manasa Gopisetty
Driver : Ramesh Kumar
Vehicle : Sedan
Pickup : Matunga
Drop : Bandra
Distance : 5.32 km
Fare : Rs. 115.73
Status : COMPLETED
=====

[Booking 2] Login_rider: Manasa Gopisetty | Vehicle=AUTO | Dadar -> Andheri
Fare Estimate : Rs.151.41
Driver_decision Assigned : Suresh Yadav
Status : COMPLETED

[Booking 3] Login_rider: Manasa Gopisetty | Vehicle=BIKE | Worli -> Parel
Fare Estimate : Rs.35.23
Driver_decision Assigned : Mahesh Patil
Status : COMPLETED

[Booking 4] change_vehicle() demo then CANCEL | SUV -> SEDAN -> CANCEL
Initial Fare (SUV) : Rs.152.97
After change_vehicle(SEDAN) : Rs.115.73
Status : CANCELLED (rider changed mind)

Demo complete. Proceeding to test suite...
=====
RIDE RECEIPT
=====
Ride ID : 1037
Rider : Gita
Driver : Ramesh
Vehicle : Sedan
Pickup : Matunga
Drop : Bandra
Distance : 5.32 km
Fare : Rs. 115.73
Status : COMPLETED
=====
```

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BLACK BOX TEST CASES (ECP + BVA) - Fare(>0) & Driver Single-Ride Constraint

TEST CASE NAME	INPUT	EXPECTED	ACTUAL	RESULT
BB-01_valid_pickup_dest_fare_positive	Login_rider=R1, Rider_pick=Matunga, Rider_dest=Bandra,	fare=115 Rs., status=PENDI	fare=115 Rs., status=PENDI	PASS
BB-02_empty_Rider_pick_name_rejected	Rider_pick name='' (empty string)	InvalidLocationException	InvalidLocationException	PASS
BB-03_login_rider_not_authenticated	Login_rider.loggedIn=false, Book Ride attempted	RideException	RideException	PASS
BB-04_BIKE_vehicle_fare_valid	Vehicle=BIKE, rate=8, Matunga->Bandra (~5km)	fare=62 Rs.	fare=62 Rs.	PASS
BB-05_AUTO_vehicle_fare_valid	Vehicle=AUTO, rate=12, Matunga->Bandra (~5km)	fare=83 Rs.	fare=83 Rs.	PASS
BB-06_SUV_vehicle_fare_valid	Vehicle=SUV, rate=25, Matunga->Bandra (~5km)	fare=152 Rs.	fare=152 Rs.	PASS
BB-07_same_pickup_dest_base_fare_only	Rider_pick==Rider_dest=Matunga, distance=0, Vehicle=AUT	fare=20 Rs.	fare=20 Rs.	PASS
BB-08_Driver_decision_on_active_ride_rejected	Driver_decision ON_RIDE, 2nd Ride details booking attem	DriverNotAvailableExceptio	DriverNotAvailableExceptio	PASS
BB-09_Driver_decision_free_ride_assigned	Driver_decision status=ONLINE, activeRideId=-1, type ma	DRIVER_ASSIGNED	DRIVER_ASSIGNED	PASS
BB-10_Driver_decision_reused_after_complete	Ride_details1 complete -> Driver_decision free -> Ride	DRIVER_ASSIGNED	DRIVER_ASSIGNED	PASS
BB-11_all_vehicle_types_Fare_positive	BIKE, AUTO, SEDAN, SUV over same Rider_pick->Rider_dest	All Fare > 0	All Fare > 0	PASS
BB-12_very_short_distance_Fare_equals_base	Rider_pick & Rider_dest 0.0001 degrees apart (~0.01 km)	fare=20.0 Rs.	fare=20.0 Rs.	PASS
BB-13_cancelRide_from_PENDING_no_driver	Ride_details status=PENDING (no driver), cancelRide() c	CANCELLED	CANCELLED	PASS
BB-14_empty_Rider_dest_name_rejected	Rider_dest name='' (empty string)	InvalidLocationException	InvalidLocationException	PASS
BB-15_Receipt_after_completeRide	completeRide() called, Receipt::print() verified, statu	COMPLETED	COMPLETED	PASS

Black Box -> Total: 15 | Passed: 15 | Failed: 0

WHITE BOX TEST CASES (Statement + Branch Coverage on assignDriver())

TEST CASE NAME	INPUT	EXPECTED	ACTUAL	RESULT
WB-01_assignDriver_available_type_match	Driver_decision ONLINE, SEDAN, activeRideId=-1	DRIVER_ASSIGNED	DRIVER_ASSIGNED	PASS
WB-02_assignDriver_driver_offline	Driver_decision status=OFFLINE, SEDAN	DriverNotAvailableExceptio	DriverNotAvailableExceptio	PASS
WB-03_assignDriver_vehicle_type_mismatch	Ride_details vehicleType=SEDAN, Driver_decision type=SU	DriverNotAvailableExceptio	DriverNotAvailableExceptio	PASS
WB-04_assignDriver_called_twice	assignDriver() called twice on same Ride_details object	RideException	RideException	PASS
WB-05_assignDriver_driver_already_on_ride	Driver_decision ON_RIDE (activeRideId!=1), 2nd Ride_de	DriverNotAvailableExceptio	DriverNotAvailableExceptio	PASS
WB-06_full_ride_lifecycle_all_statements	Enter_RideDetails->assignDriver->startRide->completeRid	COMPLETED	COMPLETED	PASS
WB-07_cancelRide_from_PENDING	Ride_details status=PENDING, cancelRide() called	CANCELLED	CANCELLED	PASS
WB-08_cancelRide_after_COMPLETED_error	Ride_details already COMPLETED, cancelRide() called	RideException	RideException	PASS
WB-09_startRide_no_driver_error	Ride_details status=PENDING, no driver, startRide() cal	RideException	RideException	PASS
WB-10_completeRide_not_started_error	Ride_details status=DRIVER_ASSIGNED, completeRide() wit	RideException	RideException	PASS
WB-11_login_rider_not_logged_in	Login_rider.loggedIn=false, Ride_details constructor ca	RideException	RideException	PASS
WB-12_Driver_decision_free_after_completeRide	completeRide() -> Driver_decision.activeRideId=-1, stat	isAvailable=true	isAvailable=true	PASS
WB-13_calculateFare_SEDAN_statement_coverage	Rider_pick=Matunga, Rider_dest=Bandra, vehicleType=SEDA	fare > 0	fare=115 Rs.	PASS
WB-14_Driver_finder_match_driver_found	Pool=[SEDAN,BIKE,AUTO,SUV], Driver_finder request BIKE	BIKE Driver_decision retur	BIKE Driver_decision retur	PASS
WB-15_Driver_finder_match_driver_nullptr	All BIKE Driver_decisions ON_RIDE -> Driver_finder retu	nullptr	nullptr	PASS

White Box -> Total: 15 | Passed: 15 | Failed: 0

OVERALL -> Total: 30 | Passed: 30 | Failed: 0 | Score: 100.0%

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STATEMENT COVERAGE

TEST CASE	NODES / STATEMENTS COVERED	STATUS
WB-01	N1,N2(no)->N3(yes)->N4->N7	COVERED
WB-02	N1,N2(no)->N3(offline)->N5->N7	COVERED
WB-03	N1,N2(no)->N3(mismatch)->N5->N7	COVERED
WB-04	N1,N2(yes)->N6->N7	COVERED
WB-05	N1,N2(no)->N3(on_ride)->N5->N7	COVERED
WB-06	N1->N4->N7 + startRide, completeRide all stmts	COVERED
WB-07	cancelRide PENDING branch	COVERED
WB-08	cancelRide COMPLETED guard	COVERED
WB-09	startRide no-driver guard	COVERED
WB-10	completeRide not-started guard	COVERED
WB-11	Ride_details constructor login guard	COVERED
WB-12	completeRide -> Driver_decision.completeRide()	COVERED
WB-13	calculateFare() SEDAN branch	COVERED
WB-14	Driver_finder::match_driver match branch	COVERED
WB-15	Driver_finder::match_driver no-match branch	COVERED

TOTAL: 7/7 CFG nodes covered = 100%

PATH COVERAGE

PATH	ROUTE	TEST CASE
PATH-1	N1->N2(driver!=null)->N6->N7	WB-04
PATH-2	N1->N2(null)->N3(avail+match)->N4->N7	WB-01
PATH-3	N1->N2(null)->N3(offline)->N5->N7	WB-02
PATH-4	N1->N2(null)->N3(type mismatch)->N5->N7	WB-03
PATH-5	startRide(DRIVER_ASSIGNED) -> ok	WB-06
PATH-6	startRide(not assigned) -> throw	WB-09
PATH-7	completeRide(IN_PROGRESS) -> ok	WB-06
PATH-8	completeRide(not started) -> throw	WB-10

TOTAL: 8/8 paths covered = 100%

CONDITION COVERAGE

COND	EXPRESSION	TRUE TC	FALSE TC	STATUS
C1	driver != nullptr	WB-04	WB-01	COVERED
C2	d->isAvailable()	WB-01	WB-02	COVERED
C3	d->getVehicleType()==vehicleType	WB-01	WB-03	COVERED
C4	status==DRIVER_ASSIGNED (startRide)	WB-06	WB-09	COVERED
C5	status==IN_PROGRESS (completeRide)	WB-06	WB-10	COVERED
C6	status==COMPLETED (cancelRide)	WB-08	WB-07	COVERED

TOTAL: 12/12 condition outcomes = 100%

PATH	ROUTE	TEST CASE
PATH-1	N1->N2(driver!=null)->N6->N7	WB-04
PATH-2	N1->N2(null)->N3(avail+match)->N4->N7	WB-01
PATH-3	N1->N2(null)->N3(offline)->N5->N7	WB-02
PATH-4	N1->N2(null)->N3(type mismatch)->N5->N7	WB-03
PATH-5	startRide(DRIVER_ASSIGNED) -> ok	WB-06
PATH-6	startRide(not assigned) -> throw	WB-09
PATH-7	completeRide(IN_PROGRESS) -> ok	WB-06
PATH-8	completeRide(not started) -> throw	WB-10

TOTAL: 8/8 paths covered = 100%

CONDITION COVERAGE

COND	EXPRESSION	TRUE TC	FALSE TC	STATUS
C1	driver != nullptr	WB-04	WB-01	COVERED
C2	d->isAvailable()	WB-01	WB-02	COVERED
C3	d->getVehicleType()==vehicleType	WB-01	WB-03	COVERED
C4	status==DRIVER_ASSIGNED (startRide)	WB-06	WB-09	COVERED
C5	status==IN_PROGRESS (completeRide)	WB-06	WB-10	COVERED
C6	status==COMPLETED (cancelRide)	WB-08	WB-07	COVERED

TOTAL: 12/12 condition outcomes = 100%

CODE COVERAGE REPORT

MODULE	LINES TOTAL	LINES HIT	COVERAGE
Ride_details	14	14	100.0%
Driver_decision	10	10	100.0%
Login_rider	5	5	100.0%
Rider_pick	6	6	100.0%
Rider_dest	5	5	100.0%
Vehicle	4	4	100.0%
Fare	4	4	100.0%
Driver_finder	4	4	100.0%
Dashboard	4	4	100.0%
Receipt	3	3	100.0%

OVERALL CODE COVERAGE: 59/59 lines = 100.0%

COVERAGE BY TEST TYPE

TEST TYPE	LINES HIT	COVERAGE
Black Box	45	81.8%
White Box	55	100.0%
Combined	55	100.0%

UNCOVERED LINES (if any)

MODULE	LINE	CODE
None	-	All lines covered